i. Cholesterol

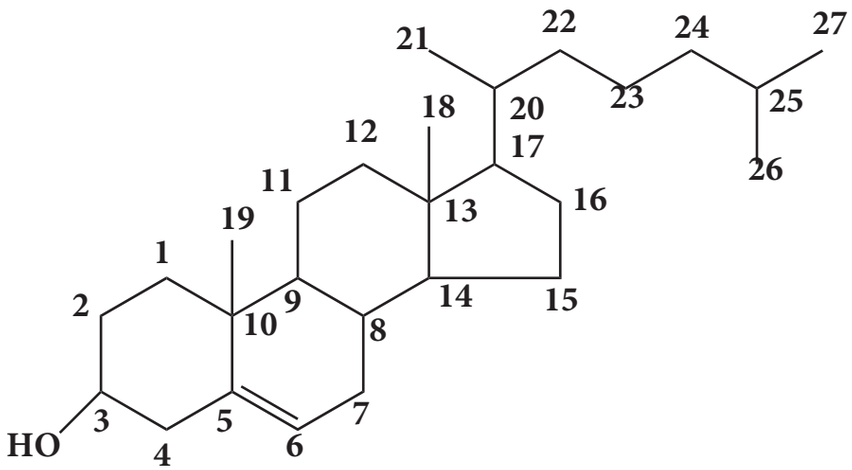
Cholesterol is widely distributed in all cells and is a major component of cell membrane and lipoproteins.

ii. Structure

Cholesterol is a C_{27} compound with molecular formula $C_{27}H_{46}O$. With a hydroxyl group at C_3 and a double bond between C_5 and C_6. An aliphatic side chain is attached at C_{17} and 5 methyl groups.

Note

Normal value of total blood cholesterol level is <200 mg/dL



Cholesterol

Figure 6.17 Structure of cholesterol

Properties of Cholestero

1. Cholesterol exists as crystals that are white, shiny and rhombic in nature.
2. It is tasteless and odourless
3. It has a high melting point of 150°C
4. It is insoluble in water and soluble in fat solvents
5. It is a poor conductor of heat and electricity and serves as an insulator. In brain, where it is present abundantly acts as an insulator against nerve impulse which are electric in nature.

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